AutoCare Platform

Project Planning, Management & Requirements Specification

PROJECT TYPE
Full-Stack Automotive Services Platform

TECHNOLOGY STACK
React, Node.js, Express.js, MongoDB

1. PROJECT OVERVIEW & SCOPE

Overview: AutoCare is a web application designed to simplify access to automotive services by connecting car owners with service centers through a centralized digital platform. Customers can explore services, compare service centers, book appointments, manage vehicle information, and track maintenance history and expenses. Service centers can manage their offerings, availability, and bookings, while administrators oversee platform operations.

Objectives: Provide a centralized discovery platform, enable service comparison (prices, location, ratings), streamline appointment scheduling, maintain digital service/expense records, and help centers manage workflow.

Scope Boundaries: Initial scope includes authentication, role-based access, vehicle management, service center listings, search/filtering, bookings, maintenance logs, and admin tooling. Integrated payments, AI diagnostics, real-time roadside assistance, and mobile apps are explicitly out of initial scope.

2. STAKEHOLDER IDENTIFICATION

Stakeholder Group	Description & Key Responsibilities
1. Car Owners (Customers)	Primary users who discover automotive services, compare service centers, book appointments, manage vehicle information, and track maintenance history and expenses.
2. Service Center Staff	Publish services, specify estimated prices, manage availability, respond to booking requests, and update service progress.
3. System Administrator	Oversee platform operations, manage users and service centers, maintain service categories, and handle reports or complaints.
4. Development Team	Analyze requirements, design and implement the system, test workflows, deploy the application, and maintain project documentation.
5. Academic Supervisor	Assess technical quality, completeness, documentation, Software Engineering practices, and fulfillment of academic objectives.

3. FUNCTIONAL REQUIREMENTS

A. Authentication & User Management

- **FR-1:** User registration with required info.
- **FR-2:** User authentication (login/logout).
- **FR-3:** Role-based access control.
- **FR-4:** Profile viewing and updates.

B. Vehicle Management

- **FR-5:** Customer vehicle record addition.
- **FR-6:** Vehicle details update and removal.
- **FR-7:** Track make, model, year, and mileage.

3. FUNCTIONAL REQUIREMENTS (CONTINUED)

C. Automotive Services & Service Centers

- **FR-8:** Display available automotive services and detailed descriptions.
- **FR-9:** Search interface for service centers by service type and name.
- **FR-10:** Filter service centers by service type, location, estimated price, and rating.
- **FR-11:** Display center profiles, working hours, active offerings, and price estimates.
- **FR-12:** Authorized center staff management of center profiles and offered services.

D. Appointment Booking

- **FR-13:** Slot booking selecting vehicle, service, center, and available time slot.
- **FR-14:** Customer booking management dashboard and status tracking.
- **FR-15:** Self-service booking cancellation based on platform timing rules.
- **FR-16:** Service center staff booking request acceptance or rejection.
- **FR-17:** Real-time booking status progress updates by center staff.
- **FR-18:** Automated prevention of double-booking/conflicting slot reservations.

E. Maintenance & Expenses

- **FR-19/20:** Maintain historical service records.
- **FR-21:** Log expenses & display summary charts.
- **FR-22:** Set up periodic maintenance reminders.

F. Reviews & Notifications

- **FR-23:** Service rating & review submission.
- **FR-24:** Booking status change notifications.
- **FR-25:** Automated maintenance reminder alerts.

G. Administration

- **FR-26:** Platform-wide user account and service center listing management.
- **FR-27:** Service category management and user complaint/report review system.
- **FR-28:** Platform activity metrics and booking audit log viewing permissions.

4. NON-FUNCTIONAL REQUIREMENTS

1. Performance: Fast API response times and scalable search filtering performance.
2. Security: Password hashing, RBAC, input validation, and secure config handling.
3. Usability: Responsive design and clear user workflows for booking and tracking.
4. Data Integrity: Strict booking state checks and data validation prior to storage.
5. Maintainability: Clean multi-tier architecture separating frontend, API, and DB.
6. Compatibility: Support for all modern web browsers across desktop and mobile screen sizes.
7. Scalability: Indexed database structures and pagination for growing data volume.
8. Availability: Robust data backing and hosting environment recovery strategies.
9. Testability: Modular design engineered to support end-to-end integration testing.
10. Privacy: Strict role-based boundaries protecting user personal data and vehicle records.

5. USER STORIES

Customer User Stories

- **US-1:** As a visitor, I want to create an account to access services.
- **US-2:** As a user, I want to log in/out securely to protect my account.
- **US-3:** As a customer, I want to manage my profile details.
- **US-4:** As a customer, I want to manage my vehicles list.
- **US-5:** As a customer, I want to record vehicle make, model, year, and mileage.
- **US-6:** As a customer, I want to browse services to find what my vehicle needs.
- **US-7:** As a customer, I want to search service centers nearby.
- **US-8:** As a customer, I want to filter centers by price, location, and rating.
- **US-9:** As a customer, I want to view center profiles and operating hours.
- **US-10:** As a customer, I want to book available slots for my vehicle.
- **US-11:** As a customer, I want to track booking statuses in real time.
- **US-12:** As a customer, I want to cancel eligible bookings when plans change.
- **US-13:** As a customer, I want to view my vehicle's maintenance history.
- **US-14:** As a customer, I want to log completed maintenance records.
- **US-15:** As a customer, I want to track auto expenses and view summary charts.
- **US-16:** As a customer, I want to configure service reminders.
- **US-17:** As a customer, I want to receive notifications for bookings and reminders.
- **US-18:** As a customer, I want to rate and review completed services.

Service Center User Stories

- **US-19:** As staff, I want to update center profile info.
- **US-20:** As staff, I want to manage service list & pricing.
- **US-21:** As staff, I want to set working hours & slot availability.
- **US-22:** As staff, I want to view incoming booking requests.
- **US-23:** As staff, I want to accept or reject requests.
- **US-24:** As staff, I want to update ongoing service progress.
- **US-25:** As staff, I want to inspect vehicle details for bookings.
- **US-26:** As staff, I want to access historical booking records.

Administrator User Stories

- **US-27:** Manage user accounts & access states.
- **US-28:** Manage center listings & verification.
- **US-29:** Maintain global service categories.
- **US-30:** Review user complaints & reports.
- **US-31:** Monitor overall system activity & logs.

QA & Development Stories

- **US-32:** Verify end-to-end workflows prior to release.
- **US-33:** Validate RBAC rules and endpoint authorization.
- **US-34:** Inspect server errors to maintain platform health.

6. CORE USE CASES

UC-01: Register an Account
Primary Actor: Visitor | **Preconditions:** Unregistered email address

Main Flow: 1. Visitor opens registration page. 2. Enters required information. 3. System validates input and checks email uniqueness. 4. System creates account securely and confirms registration.
Alternatives: Invalid input shows validation errors. Duplicate email prompts login.

UC-02: Log In to the System
Primary Actor: Registered User | **Preconditions:** Existing active account

Main Flow: 1. User submits credentials on login page. 2. System authenticates credentials. 3. System issues authentication token/session. 4. User is redirected to their role dashboard.
Alternatives: Incorrect credentials return authentication error message.

UC-03: Manage a Vehicle
Primary Actor: Customer | **Preconditions:** User is logged in

Main Flow: 1. Customer opens vehicle dashboard. 2. Customer adds or selects vehicle. 3. Enters make, model, year, and mileage. 4. System validates ownership & saves data. 5. Vehicle list refreshes.

UC-04: Search for Automotive Services
Primary Actor: Customer | **Preconditions:** Public or logged-in access

Main Flow: 1. Customer enters service keywords/location. 2. System queries active listings. 3. Customer filters by price/rating. 4. System displays matching service center profiles.

UC-05: Book an Appointment
Primary Actor: Customer & Service Center Staff | **Preconditions:** Vehicle registered, service chosen

Main Flow: 1. Customer selects vehicle, service, and center. 2. System loads available slots. 3. Customer submits slot request. 4. System verifies availability to prevent double booking. 5. Request created as 'Pending'. 6. Staff accepts/rejects request. 7. Customer is notified of decision.

UC-06: Manage and Update a Booking
Primary Actor: Service Center Staff | **Preconditions:** Authenticated staff account

Main Flow: 1. Staff opens center dashboard. 2. Staff views booking queue. 3. Staff updates booking status (Confirmed, In-Progress, Completed). 4. System updates record and notifies customer.

6. CORE USE CASES (CONTINUED)

UC-07: Cancel Appointment
Actor: Customer

Customer requests cancellation. System verifies timing rules. If eligible, status updates to 'Cancelled' and slot becomes available.

UC-08: Maintenance Records
Actor: Customer

Customer adds, edits, or deletes maintenance logs for owned vehicles. System updates historical vehicle timeline.

UC-09: Vehicle Expenses
Actor: Customer

Customer logs cost, category, and date for auto expenses. System recalculates expense analytical charts.

UC-10: Maintenance Reminders
Actor: Customer

Customer sets due dates or mileage thresholds for services. System generates notifications when milestones arrive.

UC-11: Submit a Review
Actor: Eligible Customer

After service completion, customer rates and reviews center. System validates completion status and updates rating averages.

UC-12/13: Manage Center & Slots
Actor: Center Staff

Staff configures center profile, operating hours, active services, and available booking slots while preventing overlap.

UC-14 to UC-17: Administration & System Services
Actors: Administrator / Authenticated Users

UC-14/15/16: Admins manage accounts, center listings, global service categories, resolve reported issues, and monitor analytics. **UC-17:** Secure logout invalidates authorization tokens and purges local session state.

7. GENERAL BUSINESS RULES

- Users can access only records permitted by their role and ownership.
- Customers may manage only their own vehicles, records, expenses, and bookings.
- Center staff manage only data for their authorized service centers.
- Appointment availability MUST be locked during booking to prevent double-booking.
- Only customers with verified completed bookings may submit reviews.
- State changes for bookings must follow predefined valid transitions.
- Input validation is mandatory on all data payloads prior to DB persistence.
- Automated events trigger notifications for status updates & reminders.
- Administrative actions are audited and restricted to authorized admins.
- Error handlers must protect system internals and credentials from leak.